Marmara Üniversitesi
Mühendislik Fakültesi
Elektrik-Elektronik Mühendisliği

Laboratory Manual

EE 3011
Electronics I

Experiment (2)
MOSFET IV Characteristics

Spring 2023

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MOSFET IV Characteristics

Objectives

- To study the IV curve of a MOSFET.
- To graphically find the operating point of a MOSFET circuit.
- To experimentally evaluate the threshold voltage of a MOSFET.

Pre-lab Work

Analyze the DC circuit below. Assume the transistor is in saturation and find expressions for the drain current and the drain-source voltage. Ignore channel length modulation.

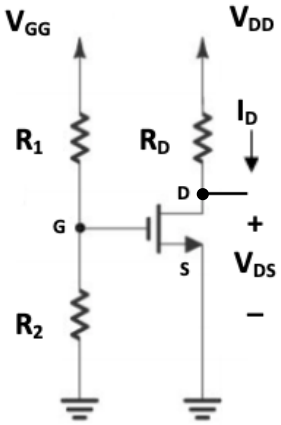


Figure 1: A MOSFET DC Circuit.

Solution

Equipment

Table 1. Equipment for experiment 6.

	Equipment	Model	Serial No.	Stock No.
1	BreadBoard Station			
2	Resistors 1 kΩ (1), 1 MΩ (2)			
3	NMOS transistor (1)	2N7000	https://www.onsemi.com/pub/Collateral/2N7000-D.PDF	
4	Multi-meter (1)			
5	Probes and cables			

Experimental Procedure

Part (1): MOSFET IV Characteristics.

1. Connect the circuit in figure 1 with R_D=1 kΩ and R_1= R_2=1 MΩ. Pay a special attention to how you connect the terminals of the MOSFET (refer to the data sheet to identify the transistor terminals).

2. Set V_{GG} to a constant voltage of +8 V while connecting V_{DD} to a variable DC voltage source.
3. The IV characteristics (I_D versus V_{DS}) is obtained by slowly varying V_{DD} (from 0 to +10 V) and measuring the voltage V_{DS} and V_{R_D} at each setting. Use table 2 to enter your results.
4. Repeat step 3 for another value of V_{GG}, say +10 V. Record your results in table 2.

Part (2): Graphical Analysis of a DC MOSFET Circuit.

1. Reconsider the circuit in figure 1, with $V_{GG} = V_{DD} = +10\,V$. Measure V_{DS} and record the results in table 3. Note that $I_D = (V_{DD} - V_{DS})/R_D$. This is the operating point of this circuit.

Part (3): MOSFET Threshold Voltage.

Produce a procedure to find the threshold voltage of the transistor using the circuit in figure 1. Write the steps below and implement them.

1.

2.

3.

4.

Results and Analysis

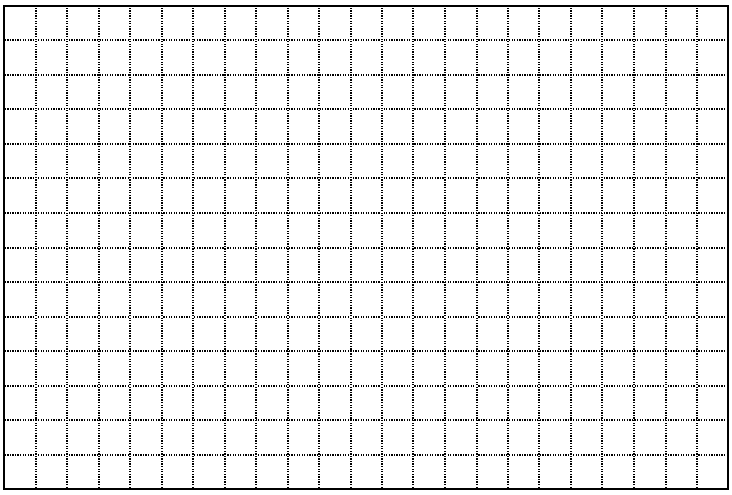
Part (1): MOSFET IV Characteristics.

1. Calculate V_{GS} (using voltage divider rule) and the current I_D (V_{R_D}/R_D) in table 2, for both values of V_{GG}.
2. Use graph 1 to sketch the IV curves (I_D versus V_{DS}) for the two V_{GS} values using the results in table 2.
3. Comment on the IV curves that you obtained. Can you identify the transistor modes of operation?

Answer:

Table 2. IV Characteristics Measurements.

	V_{GG} = 8V, V_{GS} =			V_{GG} = 10V, V_{GS} =		
V_{DD} (V)	V_{DS} (V)	V_{R_D} (V)	I_D (mA)	V_{DS} (V)	V_{R_D} (V)	I_D (mA)
0.0						
1.0						
2.0						
3.0						
4.0						
5.0						
6.0						
7.0						
8.0						
9.0						
10.0						



Graph 1: MOSFET IV Curves.

4. Is the drain current constant in the saturation region of the IV curve? Explain why or why not.

Answer:

Part (2): Graphical Analysis of a DC MOSFET Circuit.

1. The load line equation for this circuit is $I_D = \frac{V_{DD}-V_{DS}}{R_D}$. It is the line connecting the two points $(0,\frac{V_{DD}}{R_D})$ and $(V_{DD},0)$ on the IV curve.
2. Draw the load line on top of the IV curves in graph 1. Mark the intersection point of the curves.
3. The intersection of the load line and the IV curve gives the operating point of the circuit. Read the drain current (I_D) value and the drain-source voltage (V_{DS}) value from the intersection point in graph 1. Record the results in table 3.
4. The experimental value of $I_D = (V_{DD} - V_{DS})/R_D$.

Table 3. Graphical Solution Results.

V_{DD}= 10 V and V_{GG} = 10 V					
V_{DS} (V) Experimental value	V_{DS} (V) Graphical value	% Difference	I_D (mA) Experimental value	I_D (mA) Graphical value	% Difference

5. Comment on your results explaining the sources of differences.

Answers:

Part (3): MOSFET Threshold Voltage.

1. Comment on your result regarding the threshold voltage as compared to the data sheet value.
2. Comment on the procedure that you used to find the threshold voltage. How can it be improved?
3. Can you estimate the value of the constant k_n for this transistor? Show your work.

Answers:

LTspice Simulations

Include with your report LTspice simulation results of all parts of this experiment.

Conclusions and Remarks

Write below all your conclusions/remarks on this experiment.

Name		Name		Name		Name	
ID		ID		ID		ID	
Section							

2N7000G

Small Signal MOSFET 200 mAmps, 60 Volts

N–Channel TO–92

Features

- AEC Qualified
- PPAP Capable
- This is a Pb–Free Device*

MAXIMUM RATINGS

Rating	Symbol	Value	Unit
Drain Source Voltage	V_{DSS}	60	Vdc
Drain–Gate Voltage (R_{GS} = 1.0 MΩ)	V_{DGR}	60	Vdc
Gate–Source Voltage – Continuous – Non–repetitive ($t_p \le 50$ μs)	V_{GS} V_{GSM}	±20 ±40	Vdc Vpk
Drain Current – Continuous – Pulsed	I_D I_{DM}	200 500	mAdc
Total Power Dissipation @ T_C = 25°C Derate above 25°C	P_D	350 2.8	mW mW/°C
Operating and Storage Temperature Range	T_J, T_{stg}	–55 to +150	°C

THERMAL CHARACTERISTICS

Characteristic	Symbol	Max	Unit
Thermal Resistance, Junction–to–Ambient	$R_{\theta JA}$	357	°C/W
Maximum Lead Temperature for Soldering Purposes, 1/16" from case for 10 seconds	T_L	300	°C

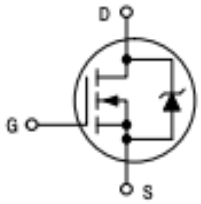
Stresses exceeding Maximum Ratings may damage the device. Maximum Ratings are stress ratings only. Functional operation above the Recommended Operating Conditions is not implied. Extended exposure to stresses above the Recommended Operating Conditions may affect device reliability.

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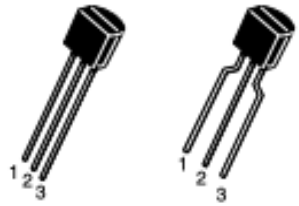
http://onsemi.com

200 mAMPS
60 VOLTS
$R_{DS(on)}$ = 5 Ω

N–Channel



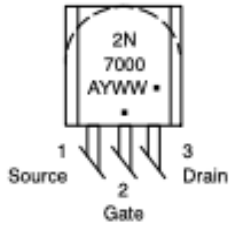
TO–92
CASE 29
STYLE 22



STRAIGHT LEAD
BULK PACK

BENT LEAD
TAPE & REEL
AMMO PACK

MARKING DIAGRAM AND PIN ASSIGNMENT



A = Assembly Location
Y = Year
WW = Work Week
• = Pb–Free Package
(Note: Microdot may be in either location)